**Topic**: Functions and transformations

**Question**: The transformation $T$ maps every vector in $\mathbb{R}^3$ to every vector in $\mathbb{R}^2$. What are the domain, codomain, and range of $T$?

**Answer choices**:

A The domain is $\mathbb{R}^3$, the codomain is $\mathbb{R}^2$, and the range is $\mathbb{R}^2$

B The domain is $\mathbb{R}^3$, the codomain is $\mathbb{R}^3$, and the range is $\mathbb{R}^3$

C The domain is $\mathbb{R}^2$, the codomain is $\mathbb{R}^3$, and the range is $\mathbb{R}^3$

D The domain is $\mathbb{R}^2$, the codomain is $\mathbb{R}^3$, and the range is $\mathbb{R}^2$

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**Solution**: A

The domain is the space $T$ is mapping *from*, so the domain is $\mathbb{R}^3$.

The codomain is the space $T$ is mapping *to*, so the codomain is $\mathbb{R}^2$.

The range is the specific set of vectors within the codomain that are being mapped to. Because every vector in $\mathbb{R}^2$ is being mapped to, the range is $\mathbb{R}^2$.

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**Topic**: Functions and transformations

**Question**: The transformation $T$ maps every vector in $\mathbb{R}^4$ to the zero vector $\vec{O}$ in $\mathbb{R}^2$. What are the domain, codomain, and range of $T$?

**Answer choices**:

A The domain is $\mathbb{R}^4$, the codomain is $\mathbb{R}^2$, and the range is $\vec{v} = (0,0)$

B The domain is $\mathbb{R}^2$, the codomain is $\mathbb{R}^4$, and the range is $\vec{v} = (0,0)$

C The domain is $\mathbb{R}^4$, the codomain is $\vec{v} = (0,0)$, and the range is $\mathbb{R}^2$

D The domain is $\mathbb{R}^2$, the codomain is $\vec{v} = (0,0)$, and the range is $\mathbb{R}^4$

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**Solution**: A

The domain is the space $T$ is mapping *from*, so the domain is $\mathbb{R}^4$.

The codomain is the space $T$ is mapping *to*, so the codomain is $\mathbb{R}^2$.

The range is the specific set of vectors within the codomain that are being mapped to, so the range is $\vec{v} = (0,0)$.

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**Topic**: Functions and transformations

**Question**: The transformation $T$ maps $\vec{a} = (1,2,-4)$ to $\vec{b} = (-3,0,4)$. What are the domain, codomain, and range of $T$?

**Answer choices**:

A The domain is $\mathbb{R}^3$, the codomain is $\vec{a} = (1,2,-4)$, and the range is $\mathbb{R}^3$

B The domain is $\mathbb{R}^3$, the codomain is $\mathbb{R}^3$, and the range is $\vec{a} = (1,2,-4)$

C The domain is $\mathbb{R}^3$, the codomain is $\vec{b} = (-3,0,4)$, and the range is $\mathbb{R}^3$

D The domain is $\mathbb{R}^3$, the codomain is $\mathbb{R}^3$, and the range is $\vec{b} = (-3,0,4)$

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**Solution**: D

The domain is the space $T$ is mapping *from*, so the domain is $\mathbb{R}^3$.

The codomain is the space $T$ is mapping *to*, so the codomain is $\mathbb{R}^3$.

The range is the specific set of vectors within the codomain that are being mapped to, so the range is $\overrightarrow{b} = (-3,0,4)$.

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**Topic**: Transformation matrices and the image of the subset

**Question**: If $\vec{a} = (-4,2)$ becomes $\vec{b}$ after undergoing a transformation by matrix $Q$, find $\vec{b}$.

$$Q = \begin{bmatrix} 11 & 1 \\ 0 & -6 \end{bmatrix}$$

**Answer choices**:

A $\vec{b} = \begin{bmatrix} 42 \\ -12 \end{bmatrix}$

B $\vec{b} = \begin{bmatrix} -42 \\ 12 \end{bmatrix}$

C $\vec{b} = \begin{bmatrix} -42 \\ -12 \end{bmatrix}$

D $\vec{b} = \begin{bmatrix} 42 \\ 12 \end{bmatrix}$

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